

Typeface structural transformations regularized using DE for personalized visual accessibility

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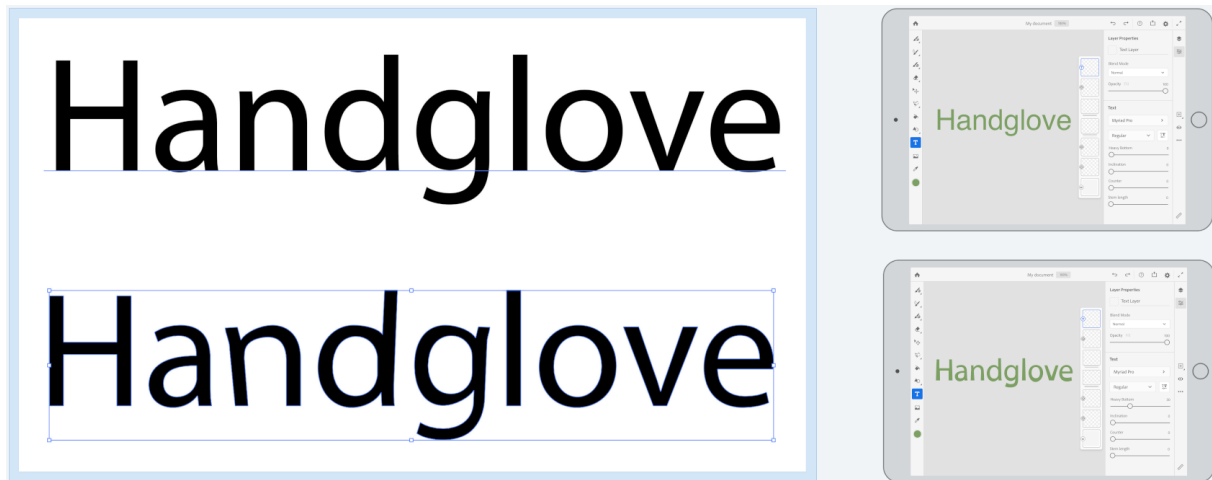


Figure 1: Applying accessibility feature transformations for personalizing visual accessibility achieved via optimal font meta-data based differential evolution. On the right side - single transformation applied by the user within an application

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**.

KEYWORDS

visual accessibility, typefaces, glyph, differential evolution

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1 INTRODUCTION

20% of world's population is suffering from Irlen syndrome, Visual discomfort (migraines), Dyslexia [dys 2011] leading to multiple cognitive traits or conditions such as letter mirroring. Studies [Gregor and et al. 2000] [Rello and et al. 2012] have shown that font lays down the foundation of an accessible visual reading experience.

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This research aims at developing a font technology with accessibility in mind, especially for platforms where personalisation is the key to enable the broad audience. Enlisted Algorithms enables every individual to apply unique transformation with controlled degree to any typeface (OTF/TTF) in vector domain ensuring easier readability. The solution applies varying degree of transformation to any typeface (referred as distinguishable fonts in the disclosure) to ease the process of reading like modifying the bottom part weight of glyphs, incline letter for better differentiation, capital letter manipulation by emphasizing the phrase end/start, modifying glyph aperture. The pipeline allows any transformation by providing candidate points and utilizing differential evolution with a typeface specific loss abiding to glyph parametric like XHeight, CapHeight, GlyphWidth, StemWidth to ensure the output typeface is still legible. Our research enables transformations in the vector domain, hence we achieve high fidelity rendering capabilities. As each transformation can either be turned on/off or else the degree of transformation can be controlled, this assistive technology can easily be presented in the user interface of the consumption tool for allowing personalisation.

2 RELATED WORK

There doesn't exist any solution that allows to manipulate textual content by personalised fonts. Although there are a few pre-established dyslexia friendly set of fonts, they can't be consumed in an application, since they only have pre-defined curves with no customization.

3 DETAILED SOLUTION AND FEATURES

3.1 Regularization pipeline: Font metadata dependent cost function based DE

As a background[Zaman and Chowdhur 2013], bezier curve could be modified by keeping shape preserved. The proposed cost function would take glyph metadata into account by taking the right parameters into the account. The output would need to be minimized so that closest shape fitting curves can be obtained while preserving the shape. For any given curve, which we intend to modify, if modified control points are (x_{mi}, y_{mi}) and original control points are (x_{ci}, y_{ci}) . Sum of these two control polygon distance can be computed as:

$$\epsilon_{dist} = \sum_{k=0}^n (y_{mk} - y_{ck})^2 + (x_{mk} - x_{ck})^2 \quad (1)$$

Now, the sum of squares error between the control points and the modified Bezier curve, can be obtained using sum of squares between their slope differences.

$$\epsilon_{slope} = \sum_{k=0}^n \left(\frac{dy_{mk}}{dx_{mk}} - \frac{dy_{ck}}{dx_{ck}} \right)^2 \quad (2)$$

If xH_{nt} can be defined as x-height normalized tolerance

$$xH_{nt} = \delta x_{height} / (\delta x_{heightMaxAllowed}) \quad (3)$$

gW_{nt} can be defined as glyph width normalized tolerance

$$gW_{nt} = \delta glyph_{width} / (\delta glyph_{widthMaxAllowed}) \quad (4)$$

Let's see how cost function can be computed with font metadata

$$f_{cost} = w_1 * \epsilon_{dist} + w_2 * \epsilon_{slope} + w_3 * xH_{nt} + w_4 * gW_{nt} \quad (5)$$

The blueprint structure uses **Step 1 as customizable** depending on the glyph modification algorithm. Rest of the steps remain same for any of the proposed glyph modification algorithms.

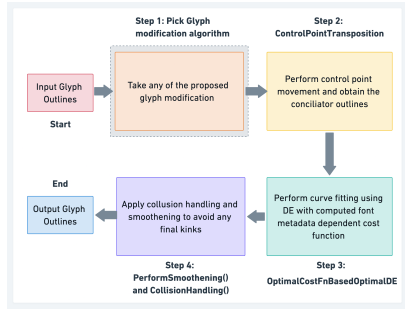


Figure 2: DE based Meta-algo for glyph modification

3.2 Method / Algo for bottom manipulation

The shown image 3 is incorporating the font attributes such as x-height, stroke width, glyph width in finalizing the cost function. This optimal DE for curve fitting can be performed to get the modified outlines for a curve segment.

3.3 Method to adjust inclination of glyphs

These subtle inclinations shown in 4 help in the sense that the different letters are easier to distinguish. For this stem inclination adjustment only X co-ordinates of control points.

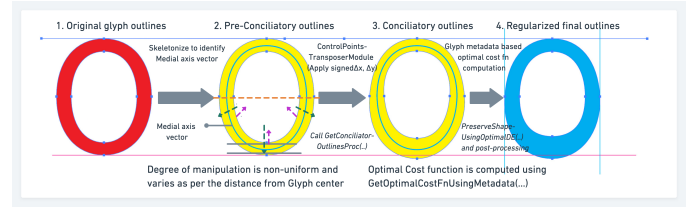


Figure 3: Bottom weight manipulation and DE meta-algo

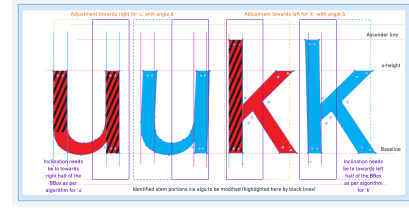


Figure 4: Stem inclination algorithm: Original vs final

3.4 Method to adjust stem length in glyphs

Degree of modification can be personalized and controlled as per the user intent (shown in 5) .

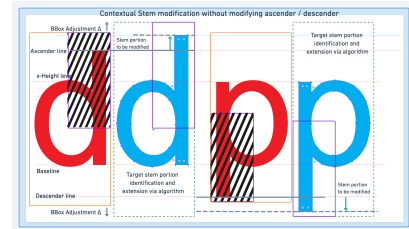


Figure 5: Stem length modifier algorithm: Original vs final

4 CONCLUSIONS

Our research enables customization for applications using categorized algorithms by modifying the rendered font glyphs in run-time. We changed the status-quo with algorithm working in vector domain, allowing precise control adjustment, the meta-algo opening up space for further research. As future work, we would like to auto-suggest settings and customization to users based on user info.

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